

SIEMENS S7-1500 · SCL FUNCTION BLOCK · MULTI-INSTANCE IDB INSIDE PPC_CONTROLLER

FB_PPC_FreqResponse

Primary frequency response (P-f droop) per ANRE Ordinul 51/2019 and 60/2024, Category D. Called from FB_PPC_Controller step ④ after FC_PPC_RampControl. Covers VAR_INPUT, VAR_OUTPUT, IDB static data, SCADA monitoring map, and ISCE test parameter table.

VERSION	IDB ACCESS PATH	CALLED FROM
0.1	"PPC_Controller".FreqResp_IDB.*	FB_PPC_Controller step ④

ANRE ARTICLES
114-115, 117, 118-120, 126-131

CORE FORMULAS

reference — computed every OB30 cycle

P-F DROOP (ART. 114-115, 118-120)

$df = f_{active} - f_{nom} \text{ [Hz]}$
 $dP_{droop} = -(2 \times Pn_{kW} \times df) / Droop_{pct} \text{ [kW]}$
 $P_{cmd_raw} = P_{setpoint_ext} + dP_{droop}$
 $P_{setpoint_final} = LIMIT(Pmin_active, P_{cmd_raw}, Pmax_active)$
Over-frequency ($df > 0$) → $dP_{droop} < 0$ → reduce P. Under-frequency ($df < 0$) → $dP_{droop} > 0$ → increase P.
Dead band: if $|df| \leq DeadBand_mHz / 1000$ then $dP_{droop} = 0$ (hard cutout — not a blend).
Example: Droop=8%, $df=+0.5 \text{ Hz}$, $Pn=48 \text{ MW}$ → $dP = -(2 \times 48000 \times 0.5) / 8 = -6\ 000 \text{ kW}$

DYNAMIC PMIN / PMAX BAND (ISCE NOTE 1)

$dP_{at_200mHz} = (2 \times Pn_{kW} \times 0.200) / Droop_{pct} \text{ [kW headroom]}$
 $Pmin_active = Pmin_stab + dP_{at_200mHz}$
 $Pmax_active = Pmax_disp - dP_{at_200mHz}$
Reserves headroom so the full ±200 mHz droop response always fits without saturation.
Recomputed every cycle — $Pmax_disp$ tracks irradiance via InverterMonitor PmaxPlant.
Example: Droop=8%, $Pn=48 \text{ MW}$, $Pmax_disp=48\ 000 \text{ kW}$ → $dP_{at_200mHz} = 2\ 400 \text{ kW}$ → $Pmax_active = 45\ 600 \text{ kW}$

VAR_INPUT 15 parameters — passed by FB_PPC_Controller from DB39				
PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST V
f_meas	REAL	Grid-measured frequency in Hz from the 4–20 mA analog input. Active source when Frequency_Source_Sel = FALSE (normal operation).	47.5 ... 52.0 Hz Nominal: 50.0 Hz	50.0 — no 50.3 — mi 49.7 — mi
Freq_Test_Override	REAL	ISCE test bench frequency (second 4–20 mA channel, 47.5–52.0 Hz). Active when Frequency_Source_Sel = TRUE. Written from SCADA to inject frequency steps during the ANRE campaign.	47.5 ... 52.0 Hz	50.0 idle 50.2 +200 50.5 +500 51.6 trip 47.4 UFRT
Frequency_Source_Sel	BOOL	Selects active frequency source. FALSE = f_meas (real grid, normal operation). TRUE = Freq_Test_Override (ISCE test bench). SCADA-writable — switch without code reload.	FALSE = f_meas TRUE = test bench	ISCE: TRUE Normal: F
f_nom	REAL	Nominal grid frequency in Hz. Used to compute $df = f_{active} - f_{nom}$. Set once at commissioning; do not change during ISCE unless instructed.	50.0 Hz (fixed)	50.0
Pn_MW	REAL	Plant nominal rated active power in MW. Used as the droop gain base: $Pn_{kW} = Pn_{MW} \times 1000$. SCADA-configurable DB parameter — stored in DB39.	48.0 MW (CEF Tandarei) Do not change during ISCE	48.0

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST V
Droop_pct	REAL	Primary frequency response droop in %. ISCE Tests 1–4 require verification at both 8% and 10%. SCADA-writable — change between test steps without code reload.	2.0 ... 12.0 % ISCE: 8.0 and 10.0	Tests 1–2 Tests 3–4
DeadBand_mHz	REAL	Frequency dead band in mHz. Hard cutout: $ df \leq \text{DeadBand}/1000 \rightarrow dP_{\text{droop}} = 0$. Not a soft blend — the test explicitly verifies a hard boundary. Set to 0 for Art.117 fine-response sub-test to verify response at $df = 50$ mHz.	0 ... 500 mHz Default: 200 mHz	Normal: 200 Art.117 sub-test: 0.0
P_setpoint_ext	REAL	Upstream P command from FC_PPC_RampControl output (Ramps_Pcmd), in kW. The droop correction is added on top of this. Represents the AGC command after rate-limiting.	0.0 ... 48 000 kW	Wired by FB_PPC_Controller — not set directly
Pmin_stab	REAL	Minimum stable power in kW — the physical lower output limit of the plant, below which it cannot operate. For PV plants this is typically 0. Added to $dP_{\text{at_200mHz}}$ to form $P_{\text{min_active}}$.	0.0 ... 5 000 kW Typical PV: 0.0	0.0
Pmax_disp	REAL	Available plant power in kW — output of FC_PPC_InverterMonitor (P_{maxPlant}). Irradiance-limited; changes every	0.0 ... 48 000 kW	In Sim_Model: 000 (10 × WRt)

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST V
		cycle as WAvail changes. Drives Pmax_active for the dynamic band.		
OFRT_Trip_Hz	REAL	Over-frequency ride-through trip threshold in Hz (Art. 118–120). When f_active exceeds this, Trip_Latch is set immediately and P_setpoint_final goes to 0.	51.0 ... 52.0 Hz Default: 51.5 Hz	Keep default 51.5 Trip test 51.6
UFRT_Trip_Hz	REAL	Under-frequency ride-through trip threshold in Hz (Art. 118–120). When f_active drops below this, Trip_Latch is set immediately.	47.0 ... 48.0 Hz Default: 47.5 Hz	Keep default 47.5 Trip test 47.4
Reconnect_Enable	BOOL	Operator/SCADA command to clear the trip latch. Only the rising edge (FALSE→TRUE) acts. Latch clears only if f_active is simultaneously back inside the [UFRT, OFRT] band. Wire from HMI push-button or SCADA momentary bit.	Momentary TRUE pulse (hold or hold-FALSE ok)	After trip TRUE when normal
Plant_Mode	INT	Current plant operating mode from FC_PPC_ModeManager. When Plant_Mode = 2 (FALLBACK), this block zeros P_setpoint_final immediately regardless of frequency, same as a trip condition.	0 = LOCAL 1 = REMOTE_IEC 2 = FALLBACK	Normal: 1 FB test: 2

VAR_OUTPUT 7 parameters — mirrored to DB39 by FB_PPC_Controller for SCADA access

PARAMETER	TYPE	DESCRIPTION	NORMAL / ALARM VALUES	DB39 MIRROR
P_setpoint_final	REAL	Active power setpoint after droop correction and dynamic-band clamp, in kW. Fed to FC_PPC_PowerDistribution as the inverter distribution input. This is the value actually sent to inverters — the primary output of the block.	0.0 ... Pmax_active = 0 during trip / FALLBACK	Not in DB39 as WSpt active
dP_droop	REAL	Droop correction term only, in kW. Diagnostic — the difference between P_setpoint_ext (AGC command) and P_setpoint_final caused by frequency deviation. Zero inside dead band. SCADA	Negative: over-freq Zero: in dead band Positive: under-freq	"PPC_Controller"
Pmin_active	REAL	Current dynamic lower bound in kW = Pmin_stab + dP_at_200mHz. P_setpoint_final is never sent below this. Changes when Droop_pct changes. SCADA	0.0 ... ~5 000 kW Typical: 2 400 kW at 8% droop, 48 MW	"PPC_Controller"
Pmax_active	REAL	Current dynamic upper bound in kW = Pmax_disp – dP_at_200mHz. P_setpoint_final is never sent above this. Tracks irradiance every cycle. SCADA	0.0 ... 48 000 kW Typical: 45 600 kW at full sun, 8% droop	"PPC_Controller"
Trip_FreqFault	BOOL	Latched TRUE when f_active crossed OFRT or UFRT threshold. While TRUE: P_setpoint_final = 0 and FB_PPC_Controller	FALSE = normal TRUE = frequency fault latched	"PPC_Controller"

PARAMETER	TYPE	DESCRIPTION	NORMAL / ALARM VALUES	DB39 MIRROR
		also resets Ramps_Pcmd = 0. Does not open a breaker — only zeros inverter setpoints. Clears on Reconnect_Enable rising edge with f back in band. SCADA ALARM		
Reconnecting	BOOL	TRUE while recovering from a trip: Trip_Latch is cleared but P_setpoint_final has not yet reached 95% of Pmax_active. Drives the reconnect timer. SCADA	FALSE = normal / tripped TRUE = ramp-up after trip	"PPC_Controller".Reconnecting
Reconnect_Timer_s	REAL	Seconds elapsed since the trip latch cleared until P reached 95% Pmax_active. Required for ISCE Test 7 reconnect timing measurement. Value freezes when Reconnecting goes FALSE — read it at that moment for the test record. ISCE TEST 7	0.0 s → freezes at end of reconnect Max timer: 3 600 s	"PPC_Controller".Reconnect_Timer_s

IDB STATIC VARIABLES (VAR) retained between OB30 cycles — accessible as "PPC_Controller".FreqResp_IDB.*

These variables live in the **PPC_Controller instance DB** under the sub-structure "PPC_Controller".FreqResp_IDB (multi-instance). Access them in the Watch table or via SCADA by navigating to that path. They are **not** mirrored to DB39 automatically — read directly from the IDB.

VARIABLE	TYPE	DESCRIPTION & SIGNIFICANCE	NORMAL STATE	HOW TO ACCESS
Trip_Latch	BOOL IDB	Internal latched trip flag. Set by any frequency exceedance (OFRT or UFRT), cleared only by a rising edge on Reconnect_Enable while f is in band.	FALSE = no trip TRUE = latched trip	"PPC_Controller".Trip_Latch

VARIABLE	TYPE	DESCRIPTION & SIGNIFICANCE	NORMAL STATE	HOV
		Same value as VAR_OUTPUT Trip_FreqFault — Trip_FreqFault is just a copy of this. Watch Trip_Latch directly in IDB if DB39 mirror is not wired yet. WRITABLE FOR TEST Can be forced FALSE in Watch table to manually clear a simulated trip without going through the Reconnect_Enable sequence.		
Reconnect_TON	TON IDB	TON timer instance measuring the reconnect duration. IN = Reconnecting output (TRUE while ramp-up in progress). PT = T#3600S (1-hour max). ET is the elapsed time. ET converted to seconds gives Reconnect_Timer_s. Read ET directly if sub-second resolution is needed beyond what REAL conversion provides.	ET = T#0S when not reconnecting Q = FALSE normally	"PP .Fr .Re
Reconnect_LastEnable	BOOL IDB	Previous-cycle value of Reconnect_Enable. Used for rising-edge detection: Rising = Reconnect_Enable AND NOT Reconnect_LastEnable. No action needed from SCADA — purely internal edge-detection memory. Visible in Watch table for debugging if the reconnect sequence seems unresponsive.	Follows Reconnect_Enable with 1-cycle lag	"PP .Fr .Re

SCADA MONITORING MAP

minimum set of tags for the ANRE ISCE commissioning campaign

ISCE TESTS 1-4

P-f Droop Verification

Frequency_Source_Sel	Set TRUE for test bench
Freq_Test_Override	Inject step: 50.2 → 50.5 → 51.0
Droop_pct (DB39)	Set 8.0 then 10.0
dP_droop (DB39)	Record — verify formula
P_setpoint_final	Verify = P_ext + dP_droop
FreqResp_Pmin/Pmax (DB39)	Verify dynamic band

ISCE TEST — ART. 117

Fine Response / Dead Band

DeadBand_mHz (DB39)	Set 0 (disable dead band)
Freq_Test_Override	Inject 50.05 (+50 mHz)
dP_droop (DB39)	Must be non-zero at 50 mHz
DeadBand_mHz (DB39)	Restore 200 after test

ISCE TEST 7

Reconnect Timing

Freq_Test_Override	Inject 51.6 to trip
Trip_FreqFault (DB39)	Verify latches TRUE
Freq_Test_Override	Return to 50.0
Reconnect_Enable (DB39)	Pulse TRUE — starts timer
Reconnecting (DB39)	Verify TRUE during ramp
Reconnect_Timer_s (DB39)	Read when Reconnecting→FALSE

CONTINUOUS MONITORING

Normal Operation

Trip_FreqFault (DB39)	Alarm if TRUE
dP_droop (DB39)	Trend — frequency event history
FreqResp_Pmax (DB39)	Trend — headroom vs irradiance
Reconnecting (DB39)	Alarm if unexpectedly TRUE
Frequency_Source_Sel	Alarm if TRUE in normal ops

TRIP — OFRT OVER-FREQUENCY

f > 51.5 Hz behaviour

Trip_Latch (IDB)	Latches TRUE immediately
P_setpoint_final	→ 0 kW (no ramp)
Ramps_Pcmd (DB39)	Reset to 0 by Controller
Trip_FreqFault (DB39)	→ TRUE
Reconnect_Timer_s	Starts on Reconnect_Enable pulse

WATCH TABLE — DEBUG

IDB Direct Access

.FreqResp_IDB.Trip_Latch	Force FALSE to clear trip in test
.FreqResp_IDB.Reconnect_Timer	Raw timer value (TIME type)
.FreqResp_IDB.Reconnect_Latch	Check edge detection is working
Prefix all paths with "PPC_Controller"	